

Regime-Gated Generative Models for Probabilistic Financial Forecasting

A chronological comparison of self-supervised Transformers, Tail-GAN and stress-shrunk GOM against an EWMA Student-t benchmark

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Research portfolio paper

Abstract. This study develops and evaluates a baseline-anchored framework for probabilistic forecasting of daily SPY returns at one- and five-day horizons. The framework combines a frozen exponentially weighted Student-t reference model, a conservatively admitted statistical expert pool, a self-supervised multi-resolution patch Transformer, and two constrained scenario generators: a stabilized Tail-GAN and a stress-shrunk generative objective model (GOM). Machine-learning forecasts enter the final distribution only through bounded residual corrections. Their activation is determined before test evaluation by a chronological gate that separates selection, tuning and confirmation data, requires consistent episode and regime evidence, applies a circular block-bootstrap lower-bound test, constrains risk-calibration deterioration, and rejects concentrated seed ensembles. Across three development folds, no model was promoted in 2020. In 2022, GOM and Tail-GAN reduced mean pinball loss relative to the frozen EWMA Student-t forecast by 0.408% and 0.285%, respectively; both common-origin loss-difference intervals were entirely favourable. Tail-GAN also produced a 0.203% gain in 2024, although its descriptive interval included zero. The results show that small but repeatable improvements can be obtained from regime-conditioned residual learning without replacing a robust statistical forecast. They also show that better aggregate quantile scores do not by themselves resolve Value-at-Risk exception clustering, particularly at the five-day horizon.

Keywords: probabilistic forecasting; generative adversarial networks; quantile regression; regime switching; self-supervised learning; Value-at-Risk; financial time series

JEL classifications: C22, C45, C53, G17

Summary of contributions

1. A dual-reference statistical champion that preserves the frozen EWMA Student-t forecast unless an adaptive heavy-tail expert pool passes disjoint chronological checks.
2. A target-relevant patch Transformer pretrained on masked histories, future distribution descriptors and regime features, with a zero-initialized quantile-residual head.
3. Stable factor-scale Tail-GAN and stress-shrunk GOM generators whose influence is bounded by model-family and regime.
4. A probabilistic promotion gate that combines seed regularization, confidence-weighted regime routing, multi-episode confirmation, block bootstrap evidence and risk-calibration constraints.
5. A formal chronological evaluation in which two models outperform EWMA in the 2022 fold while rejected forecasts remain exactly equal to the statistical benchmark.

1. Introduction

Financial forecasting is difficult not only because expected returns are small, but because the conditional distribution changes across volatility, liquidity and macroeconomic regimes. A point forecast can therefore be accurate on average while understating downside risk, and a flexible neural model can appear successful on a later sample even when its pre-test validation evidence is weak. These features motivate a distributional forecasting problem in which the full conditional quantile surface is evaluated chronologically and the machine-learning model is prevented from displacing a stronger statistical forecast without prior evidence.

Generative adversarial networks (GANs) offer an implicit route to conditional distribution learning [1]. Conditional forecasting architectures such as ForGAN [6] and finance-specific objectives such as Fin-GAN [7] demonstrate how adversarial models can be adapted from unconditional sample generation to supervised financial prediction. Tail-GAN further links scenario generation to elicitable tail-risk functionals [8]. Nevertheless, GAN optimization is unstable, the relationship between validation and test regimes can be weak, and a favourable test score does not establish that the model could have been selected without observing that score. The present study addresses those concerns by anchoring every neural forecast to a fixed EWMA Student-t model and by treating model promotion as a separate chronological decision problem.

The experiment uses three development folds centred on the COVID-19 crisis, the inflation and rate-shock regime, and the more recent 2024 market. The 2025 period is not used in any model fitting, selection or development comparison. This separation allows the paper to ask a narrower and more defensible question: can a constrained neural residual improve a robust benchmark under a predeclared promotion rule?

RiskGraph v1.7: baseline-anchored residual learning and formal promotion

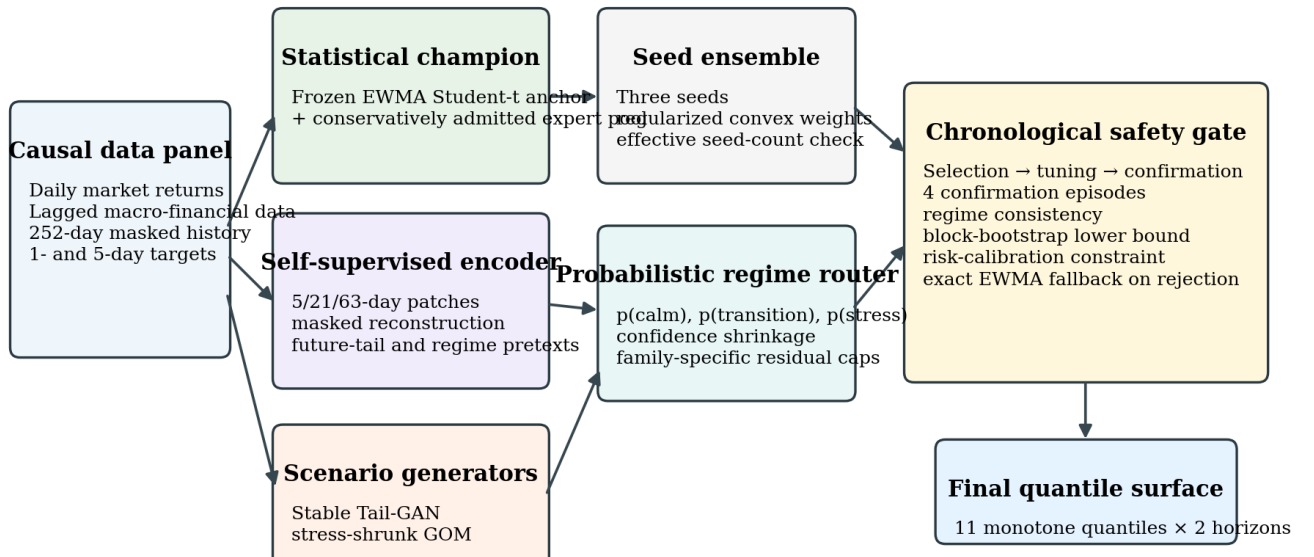


Figure 1. Architecture of the RiskGraph v1.7 forecasting system. Statistical and neural components are trained causally; the machine-learning families affect the final forecast only through bounded residual corrections selected by the chronological safety gate.

2. Related literature

2.1 Generative models for financial time series

The original GAN formulates distribution learning as a two-player minimax game between a generator and a discriminator [1]. Wasserstein GANs replace a divergence-based objective with an approximation to the Wasserstein-1 distance [2], while gradient penalties [3] and spectral normalization [4] improve critic regularity. These ideas are relevant for financial data because tail observations are sparse and unstable gradients can cause the generator to ignore important regions or collapse toward a narrow set of outcomes.

ForGAN conditions a recurrent generator and discriminator on recent history to obtain probabilistic one-step forecasts [6]. Fin-GAN extends that construction with economically motivated loss terms and evaluates generated distributions through trading-oriented criteria [7]. Tail-GAN instead targets portfolio tail risk by exploiting the joint elicibility of Value-at-Risk and Expected Shortfall [8]. TimeGAN [9] and COT-GAN [10] provide broader approaches for sequential generation, respectively through supervised latent dynamics and causal optimal transport. The present work differs from these models in its primary objective: it estimates a quantile surface around a fixed statistical forecast, and promotion is conditional on chronological gate evidence rather than on generator loss alone.

2.2 Transformers and self-supervised representation learning

Self-attention provides a parallel mechanism for relating distant observations [11]. PatchTST shows that segmenting long time series into patch tokens can improve computational efficiency, preserve local information and support self-supervised pretraining [12]. The structured model in this study follows the patching principle but introduces multi-resolution patches, explicit missingness channels and baseline-relative quantile parameters. Its pretraining tasks are designed to extract information about volatility, downside movement and regime state rather than to reconstruct the target series alone.

2.3 Quantile scoring and risk backtesting

Quantile forecasts are evaluated by the asymmetric pinball loss introduced by quantile regression [13]. This score is strictly consistent for a conditional quantile and therefore provides an appropriate objective for the complete distributional surface. The lower 5% quantile is also interpreted as a Value-at-Risk forecast. Unconditional coverage is examined with the likelihood-ratio test of Kupiec [14], and serial independence of exceedances with the conditional-coverage framework of Christoffersen [15]. Because financial loss differences are serially dependent, confirmation evidence is supplemented by circular block resampling [16].

3. Forecasting problem and notation

Let P_t denote the adjusted SPY closing price and r_t the one-day log return. For forecast origin t and horizon $h \in \{1,5\}$, the target is the cumulative return $y_{t,h}$.

$$r_t = \log(P_t / P_{t-1}), \quad y_{t,h} = \sum_{j=1}^h r_{t+j}. \quad (1)$$

The model produces $Q(t,h,\tau)$, an estimate of the conditional τ -quantile of $y_{t,h}$ for eleven probability levels $\tau \in \{0.01, 0.025, 0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95, 0.975, 0.99\}$. Quantile crossing is removed by monotone rearrangement at each origin and horizon.

For residual $u = y - q$, the pinball loss is

$$\rho_\tau(u) = u[\tau - 1\{u < 0\}], \quad L(Q) = (1 / NHK) \sum_t \sum_h \sum_\tau \rho_\tau(y_{t,h} - Q(t,h,\tau)). \quad (2)$$

A candidate is reported as an improvement of $\Delta\%$ relative to reference B when

$$\Delta(Q;B) = 100 \times [L(B) - L(Q)] / L(B). \quad (3)$$

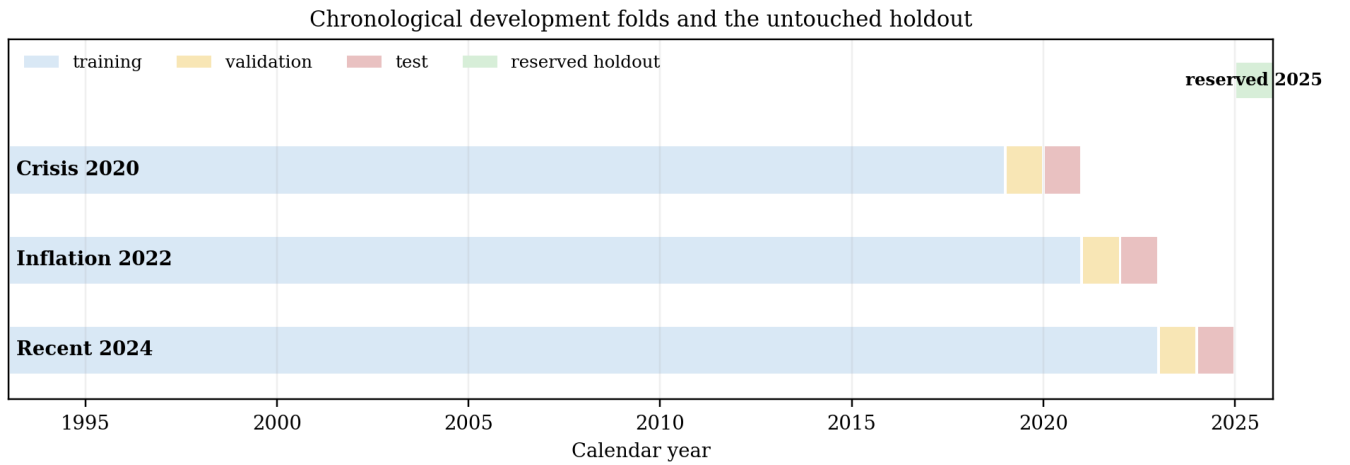


Figure 2. Chronological training, validation and test periods. The green 2025 interval is excluded from the development analysis.

4. Model construction

4.1 Frozen EWMA Student-t anchor

The fixed benchmark is an exponentially weighted location-scale model with Student-t innovations. Its mean and variance states evolve as

$$\mu_t = \lambda_\mu \mu_{t-1} + (1-\lambda_\mu) r_t, \quad \sigma_t^2 = \lambda_\sigma \sigma_{t-1}^2 + (1-\lambda_\sigma)(r_t - \mu_t)^2, \quad (4)$$

where $\lambda_\mu = 0.985$ and $\lambda_\sigma = 0.94$. The degrees of freedom ν are fitted using training data only. The h -step anchor quantile uses square-root-of-time scaling:

$$Q_A(t,h,\tau) = h \mu_t + \sqrt{h} \sigma_t T_{\nu-1}^{-1}(\tau). \quad (5)$$

The anchor is deliberately simple. It adapts quickly to volatility and supplies heavy tails, while its low dimensionality reduces estimation variance. All later components are evaluated relative to this same forecast.

4.2 Statistical champion

An adaptive pool contains fast, standard, medium, slow, leverage-sensitive and asymmetric heavy-tail EWMA experts. For horizon h , non-negative weights $a_{\{h,j\}}$ sum to one, yielding $Q_P = \sum_j a_{\{h,j\}} Q_j$. The pool is fitted on an early training segment and then compared with the frozen anchor on later, disjoint training segments. A candidate champion is

$$Q_S = M\{m + s_h[(Q_A + \beta_h(Q_P - Q_A)) - m] + \delta_{\{h,\tau\}}\}, \quad (6)$$

where m is the median quantile, β_h is the anchor-pool blend, s_h rescales the spread, $\delta_{\{h,\tau\}}$ is a shrunk quantile offset and M denotes monotone rearrangement. The adaptive pool is admitted only if confirmation improvement, episode consistency and a one-sided block-bootstrap lower bound satisfy the predeclared conditions. In all three folds of this experiment, $\beta_h = 0$ for both horizons; hence Q_S is exactly Q_A .

4.3 Target-relevant self-supervised patch Transformer

The structured model receives 252 daily observations with value and availability channels. Histories are standardized within each sample, segmented into 5-, 21- and 63-day patches, and projected to channel representations. Learned attention weights pool channels within each patch before the resulting temporal tokens enter a three-layer Transformer encoder. Pretraining combines masked-history reconstruction, regime prediction, future-distribution descriptors and contrastive agreement between independently corrupted views.

$$L_{SSL} = L_{mask} + 0.20 L_{regime} + 0.35 L_{future} + 0.10 L_{contrastive}. \quad (7)$$

The supervised head predicts four bounded parameters per horizon: a location shift, lower-tail scale, upper-tail scale and curvature. If q_τ is a baseline quantile, m the median and R the 1%-99% range, the transformed quantile is

$$Q_T(\tau) = m + \ell R + s(\tau) \cdot \text{sign}(q_\tau - m) \cdot R / [q_\tau - m / R]^p. \quad (8)$$

The final layer is initialized to zero, so the initial network reproduces the statistical champion exactly. A one-sided baseline guard penalizes increases in supervised pinball loss, thereby discouraging unsupported departures from the anchor.

4.4 Stable Tail-GAN and stress-shrunk GOM

The scenario models retain an EWMA factor-covariance backbone rather than generating unrestricted returns. The generator predicts bounded origin-level controls for common-factor scale, idiosyncratic scale, conditional drift and downside asymmetry. A schematic scenario is

$$\tilde{r}_{\{t+1:t+H\}} = \mu_t + d_t + B_t D_f(s_t) z_f + D_\varepsilon(u_t) z_\varepsilon + \kappa_t \min(z, 0), \quad (9)$$

with Student-t innovations z_f and z_ε , low-rank factor loading B_t , bounded controls $(s_t, u_t, d_t, \kappa_t)$, and $H = 20$. Tail-GAN uses adversarial and tail-score objectives, whereas GOM places greater weight on marginal, correlation and stress-relevant distributional errors. Critic spectral normalization, output regularization, gradient clipping, normalized tail scores, exponential moving averages and early stopping improve stability [3,4]. GOM receives tighter parameter limits and can use at most 0.10, 0.30 and 0.70 residual activation in calm, transition and stress states.

4.5 Probabilistic regime router

A causal regime model converts recent volatility, downside variation, drawdown, market stress and cross-asset dispersion into probabilities $p_t = (p_{\text{calm}}, p_{\text{transition}}, p_{\text{stress}})$. Forward-only persistence smoothing reduces state switching. Let c_t be regime confidence, c_0 the confidence floor and w the family-specific residual weights. The activation strength is

$$\alpha_t = \text{clip}[(c_t - c_0) / (1 - c_0), 0, 1] \cdot \sum_k p_{\{t,k\}} w_k. \quad (10)$$

The routed forecast is therefore a convex residual interpolation

$$Q_G(t, h, \tau) = Q_S(t, h, \tau) + \alpha_t [Q_M(t, h, \tau) - Q_S(t, h, \tau)]. \quad (11)$$

When confidence is low or the gate rejects the family, $\alpha_t = 0$ and the output equals the statistical champion exactly.

4.6 Seed ensemble and promotion gate

Three independently trained seeds are combined with non-negative weights π_s . Weights are shrunk toward equality, capped at 0.80 and penalized for concentration. Effective seed count is

$$N_{\text{eff}} = 1 / \sum_s \pi_s^2. \quad (12)$$

The validation period is partitioned into selection, tuning and confirmation blocks. Selection chooses π and the regime residual weights; tuning provides a first untouched check; confirmation supplies the final promotion evidence. Acceptance requires positive confirmation improvement, a strictly positive one-sided circular block-bootstrap lower bound, at least three favourable confirmation episodes out of four, bounded episode and regime degradation, limited risk-calibration deterioration and $N_{\text{eff}} \geq 1.35$.

5. Algorithms

Algorithm 1. Construction of the statistical champion

Input: training returns, horizons H , quantiles T , frozen EWMA anchor A , expert bank $\{E_j\}$.

- 1 Split the final training history into pool-fit, blend-selection, calibration and confirmation blocks.
- 2 Fit convex horizon-specific expert weights on the pool-fit block.
- 3 Search anchor-pool blend β_h on the blend-selection block.
- 4 Estimate spread scale s_h and shrunk quantile offsets $\delta_{\{h,\tau\}}$ on the calibration block.
- 5 Measure confirmation improvement by chronological episodes and circular block bootstrap.
- 6 If every admission condition passes, return the calibrated blend; otherwise return A exactly.

Output: causal statistical champion Q_S .

Algorithm 2. Model training and probabilistic promotion

Input: champion forecasts Q_S , histories X , targets y , regime probabilities p , seeds $S=\{11,22,33\}$.

- 1 Train each neural seed using training data only; export raw validation and test quantile forecasts.
- 2 On validation-selection data, regularize seed weights π and search family-capped regime weights w .
- 3 Reject if selection improvement is below the declared threshold.
- 4 Evaluate the fixed (π, w) on a disjoint tuning block.
- 5 Evaluate the same choice on confirmation data; compute episode, regime, bootstrap and risk diagnostics.
- 6 Accept only if every condition passes; set $Q_G = Q_S + \alpha(p, w)(Q_M - Q_S)$.
- 7 On rejection, set $Q_G = Q_S$ exactly and serialize the baseline frame without numerical reconstruction.

Output: final monotone quantile forecast and auditable gate record.

Algorithm 3. Formal chronological evaluation

for fold in {2020 crisis, 2022 inflation, 2024 recent} do
 fit all statistical and neural components using dates no later than the fold training boundary;
 select model and gate parameters using validation dates only;
 freeze the fold decision and evaluate on common test origins;
 compare final forecasts with the statistical champion and frozen EWMA anchor;
 verify finite monotone quantiles, exact fallback, seed diversity and accepted-gate conditions;
end for
Keep 2025 outside all development procedures.

6. Experimental design

6.1 Data and chronological splits

The target series is SPY. The multivariate panel begins in January 1993 and includes broad equity indices, sector exchange-traded funds and lagged macro-financial variables such as the VIX, credit spread, Treasury yields, the policy rate, financial-condition indices and oil. Macroeconomic series are shifted by one day to reduce release-timing leakage. All models share common forecast origins, a five-day embargo and the same one- and five-day targets.

Table 1. Chronological forecasting folds.

Fold	Training end	Validation	Test	Common test dates
Crisis 2020	≤ 31 Dec 2018	2019	2020	229
Inflation 2022	≤ 31 Dec 2020	2021	2022	226
Recent 2024	≤ 31 Dec 2022	2023	2024	228
Reserved holdout	≤ 31 Dec 2023	2024	2025	not evaluated

6.2 Principal hyperparameters

Table 2. Principal design settings.

Component	Specification
Statistical anchor	EWMA mean decay 0.985; variance decay 0.94; Student-t degrees of freedom fitted on training data
Expert-pool admission	6 episodes; 1,500 bootstrap repetitions; 80% one-sided lower bound; 0.1% minimum confirmation gain
Transformer	252-day history; patches 5/21/63; d_{model} 128; 3 layers; 4 heads; 20 pretraining + 50 supervised epochs
Scenario generators	20-day paths; rank 4; 65 epochs; spectral-normalized critic; 256 scenarios per test origin
Promotion gate	45% selection, 25% tuning, remaining confirmation; 2,000 bootstrap repetitions; 4 episodes; confidence floor 0.45
Seed control	3 seeds; 15% shrinkage toward equal weights; maximum seed weight 0.80; minimum N_{eff} 1.35

6.3 Evaluation criteria

Mean pinball loss is the primary forecast score. Supporting diagnostics include median MAE and RMSE, directional accuracy, 90% and 95% interval coverage and width, interval scores, 5% VaR exception rate, the Kupiec coverage test, the Christoffersen independence test and expected-shortfall comparisons. Common-origin confidence intervals are descriptive block-bootstrap intervals; they do not participate in model selection.

6.4 Reproducibility controls

The formal matrix contains 27 supervised seed models and nine fold-family ensemble decisions. Every accepted ensemble uses the same three seeds and the same common-origin benchmark. Rejected predictions are written as exact copies of the champion forecast, avoiding floating-point discrepancies caused by serializing a reconstructed array. The source tree contains automated tests for monotonicity, causality, seed concentration, exact fallback and holdout protection.

7. Results

7.1 Formal fold comparison

Table 3. Full formal fold-family results.

Fold	Family	Raw pinball	Final pinball	vs EWMA	Decision
Crisis 2020	Patch Transformer	0.00514144	0.00518217	-0.000%	fallback
Crisis 2020	Tail-GAN	0.00506998	0.00518217	-0.000%	fallback
Crisis 2020	Stress-shrunk GOM	0.00507449	0.00518217	-0.000%	fallback
Inflation 2022	Patch Transformer	0.00412451	0.00412580	0.000%	fallback
Inflation 2022	Tail-GAN	0.00412255	0.00411403	0.285%	accepted
Inflation 2022	Stress-shrunk GOM	0.00408559	0.00410896	0.408%	accepted
Recent 2024	Patch Transformer	0.00204507	0.00204523	0.000%	fallback
Recent 2024	Tail-GAN	0.00206452	0.00204107	0.203%	accepted
Recent 2024	Stress-shrunk GOM	0.00206505	0.00204523	0.000%	fallback

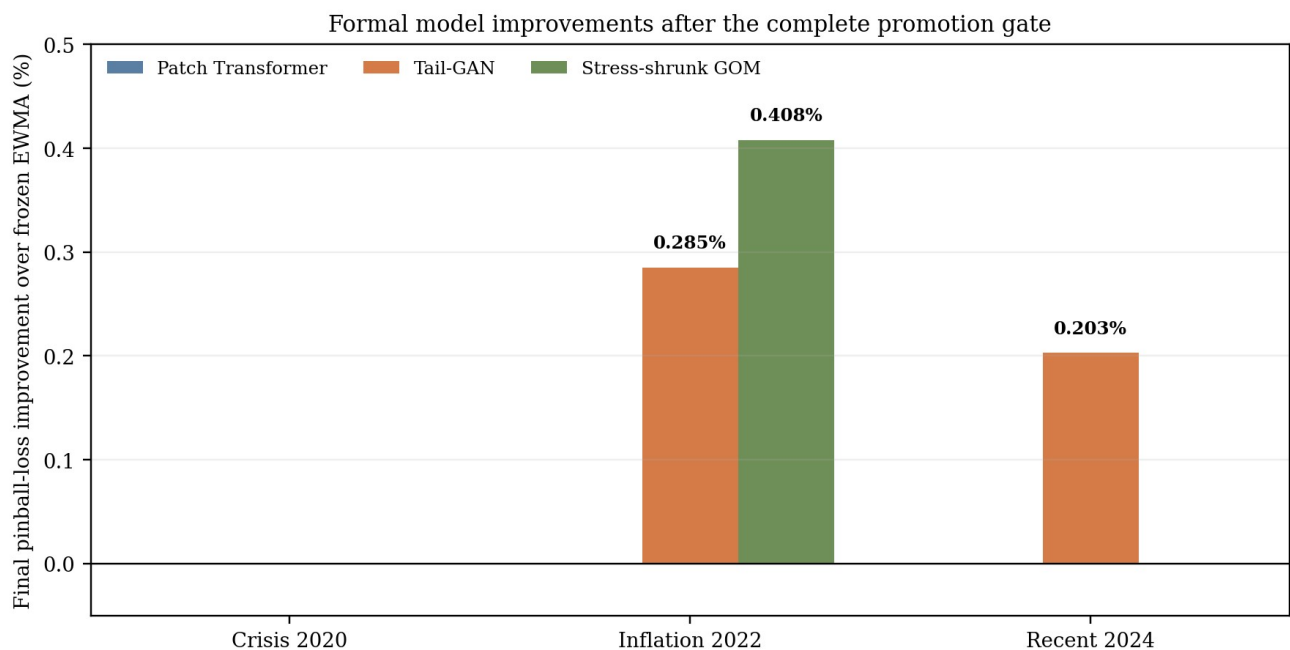


Figure 3. Formal improvements after the complete promotion gate. A zero bar denotes exact statistical fallback.

Raw test performance and the effect of the predeclared gate

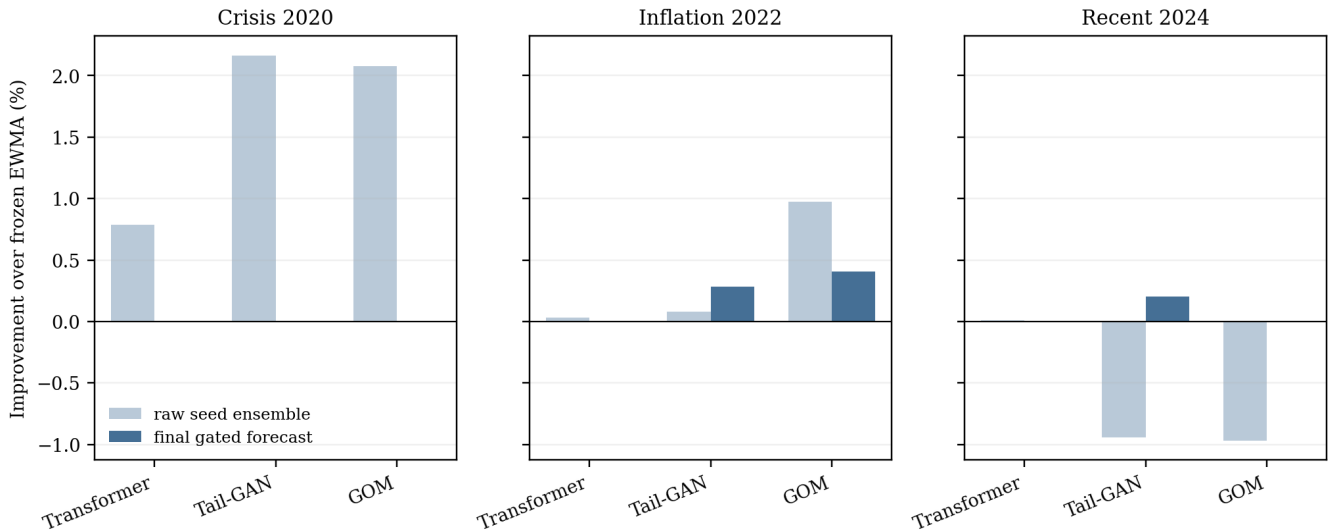


Figure 4. Raw and final improvement relative to the frozen EWMA Student-t anchor. The 2020 test sample favoured the raw generators, but pre-test evidence was insufficient for promotion.

7.2 Crisis 2020

No machine-learning family was promoted. The EWMA Student-t anchor achieved mean pinball loss 0.00518217. Raw Tail-GAN, GOM and Transformer ensembles scored better on the later test sample, but those gains were not supported by the pre-test decision rule. GOM failed the bootstrap condition, the Transformer failed the selection threshold, and Tail-GAN failed episode, regime or confirmation consistency. The formal output therefore remained the anchor for all three families.

This fold illustrates the distinction between predictive opportunity and valid model selection. The raw results suggest that the models contain crisis-relevant information; the gate result states that the information was not identifiable reliably from the preceding chronology. Ex-post acceptance would convert a model-development observation into a biased performance claim.

7.3 Inflation 2022

Two models were accepted. Stress-shrunk GOM reduced mean pinball loss from 0.00412580 to 0.00410896, a 0.408% improvement. Tail-GAN reduced the loss to 0.00411403, a 0.285% improvement. Their descriptive loss-difference intervals against the frozen anchor were $[4.58 \times 10^{-6}, 3.18 \times 10^{-5}]$ and $[3.06 \times 10^{-6}, 2.17 \times 10^{-5}]$, respectively, so both intervals lay entirely on the favourable side. Each ensemble used equal seed weights, $N_{\text{eff}} = 3$, four non-negative confirmation episodes and a positive one-sided block-bootstrap lower bound.

GOM applied regime residual weights (0.10, 0.20, 0.70) for calm, transition and stress conditions. Tail-GAN used (0.10, 0.20, 0.35). The stronger stress cap of GOM is consistent with its design as a conditional stress specialist. The raw GOM ensemble was better than the final routed forecast; the gate traded some point improvement for a bounded, more conservative activation pattern. Tail-GAN showed the opposite pattern: routing improved a relatively modest raw result.

7.4 Recent 2024

Tail-GAN was accepted with a 0.203% reduction in mean pinball loss, from 0.00204523 to 0.00204107. Validation episodes and the block-bootstrap lower bound were positive, and the three seed weights remained equal. The descriptive common-origin interval, however, was $[-3.12 \times 10^{-6}, 1.34 \times 10^{-5}]$, which includes zero. The 2024 fold therefore provides weaker test-sample evidence than the 2022 fold even though it satisfies the predeclared promotion rule.

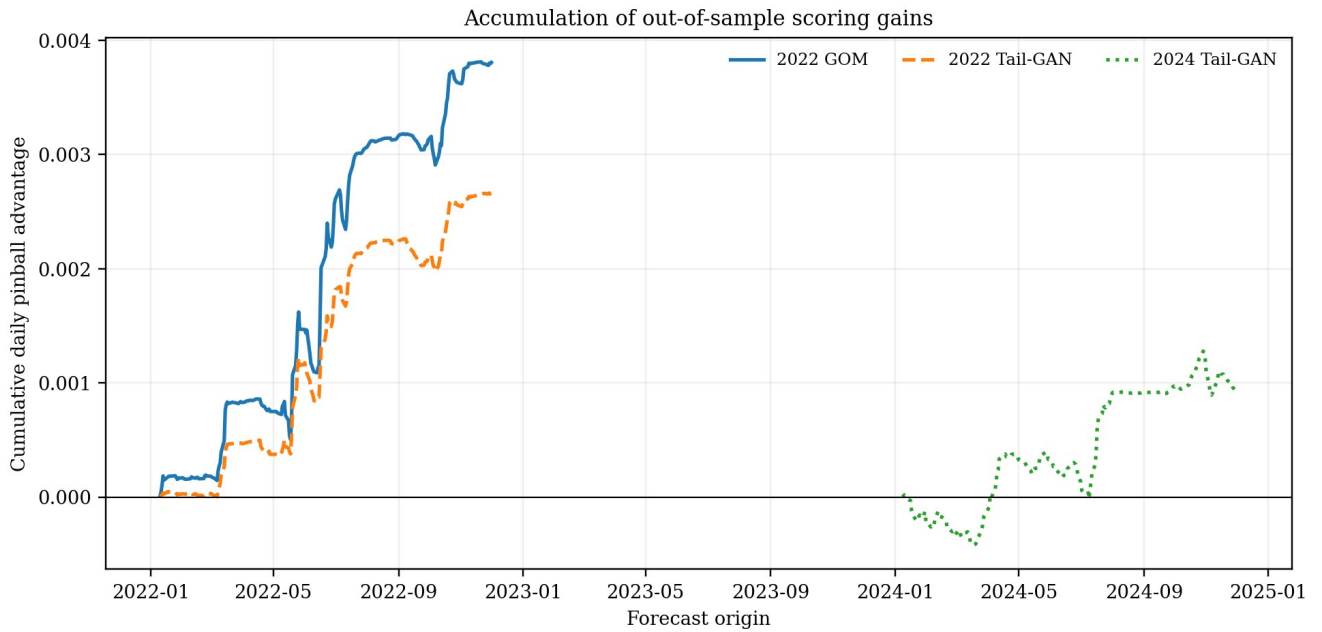


Figure 5. Cumulative daily loss advantage of the accepted forecasts. Positive slopes indicate periods in which the gated model improves on the frozen anchor.

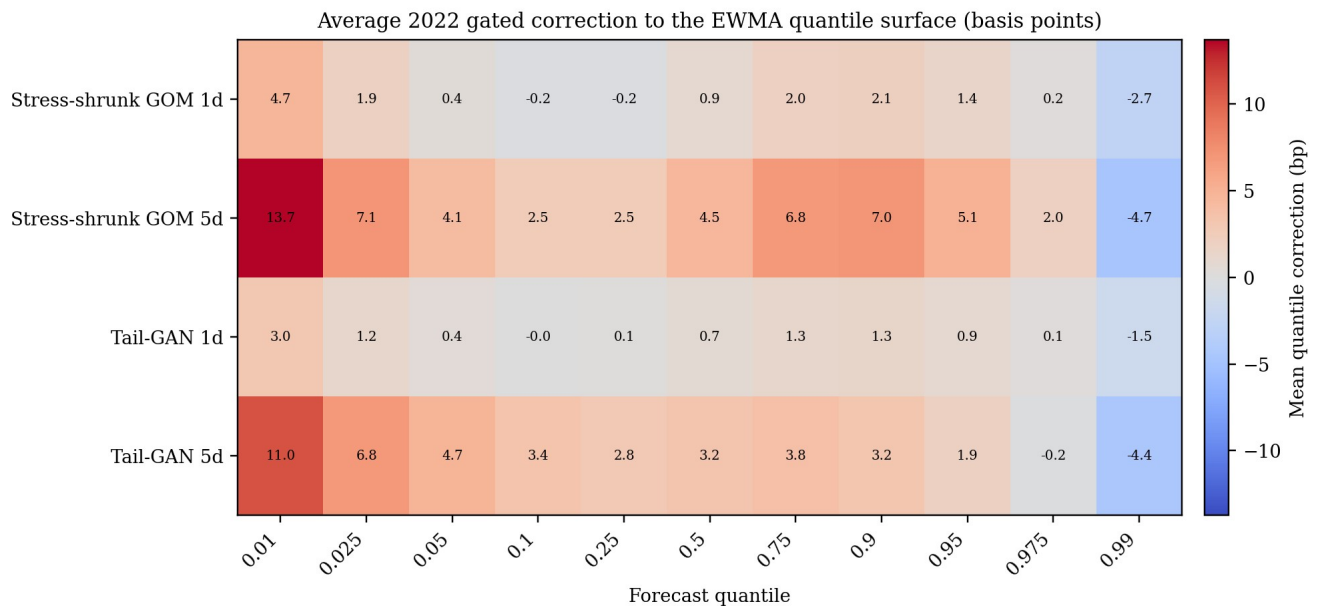


Figure 6. Mean correction applied to the 2022 EWMA quantile surface. The corrections are small relative to daily return variability and differ by horizon and tail probability.

7.5 Horizon decomposition

Table 4. Accepted models by horizon.

Fold	Model	Horizon	Pinball	Improvement	90% coverage	5% exception
Inflation 2022	Stress-shrunk GOM	1 day	0.00255224	0.117%	0.854	0.084
Inflation 2022	Stress-shrunk GOM	5 day	0.00566568	0.539%	0.854	0.093
Inflation 2022	Tail-GAN	1 day	0.00255333	0.075%	0.854	0.084
Inflation 2022	Tail-GAN	5 day	0.00567473	0.380%	0.854	0.093
Recent 2024	Tail-GAN	1 day	0.00126160	0.044%	0.882	0.079
Recent 2024	Tail-GAN	5 day	0.00282054	0.274%	0.877	0.079

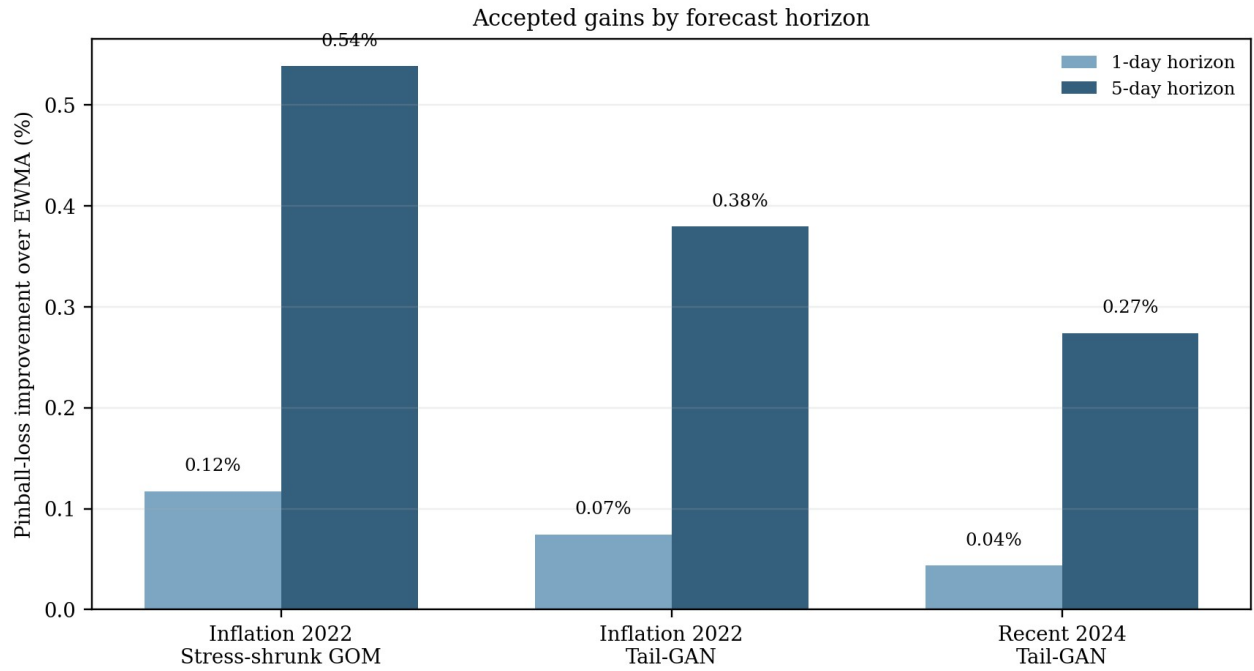


Figure 7. Horizon-specific pinball-loss improvements. The five-day component supplies the larger gain for each accepted model.

7.6 Gate evidence and regime activation

Table 5. Gate evidence for accepted ensembles.

Fold	Model	Tuning	Confirmation	Bootstrap LB	N _{eff}	Calm/Trans./Stress
Inflation 2022	Stress-shrunk GOM	0.228%	0.754%	1.04e-05	3.0	0.10/0.20/0.70
Inflation 2022	Tail-GAN	0.264%	0.619%	8.58e-06	3.0	0.10/0.20/0.35
Recent 2024	Tail-GAN	0.613%	0.197%	3.32e-06	3.0	0.10/0.20/0.35

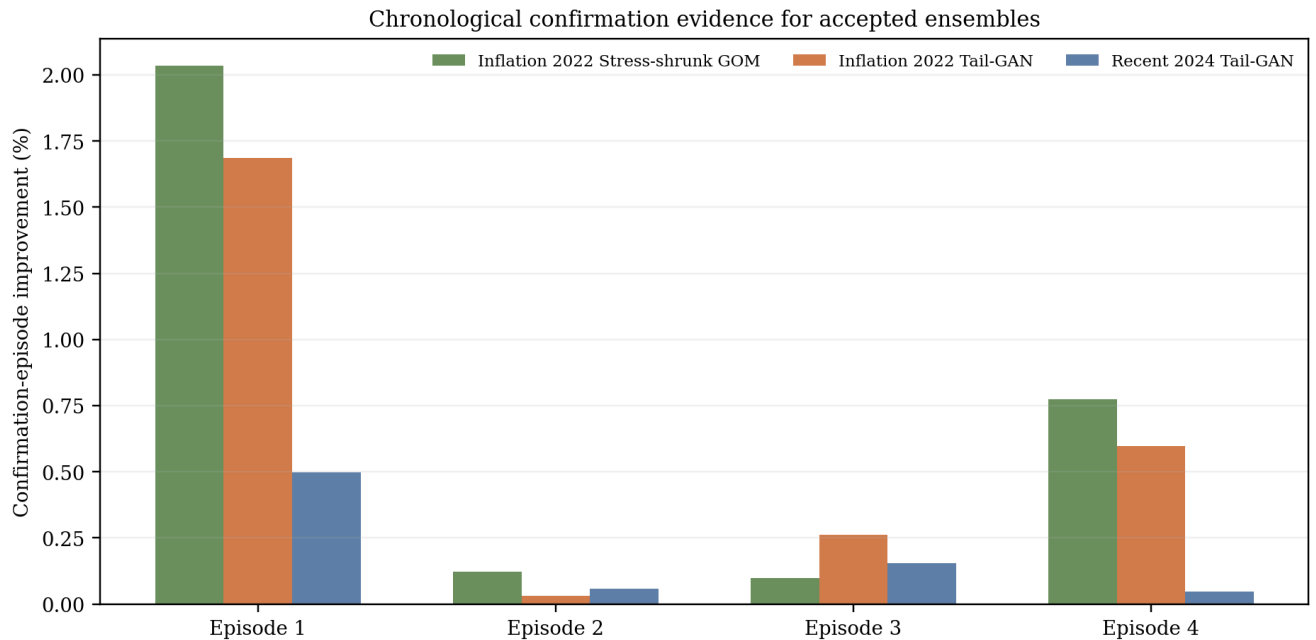


Figure 8. Improvements in the four chronological confirmation episodes. Every accepted model satisfies the required positive-episode fraction.

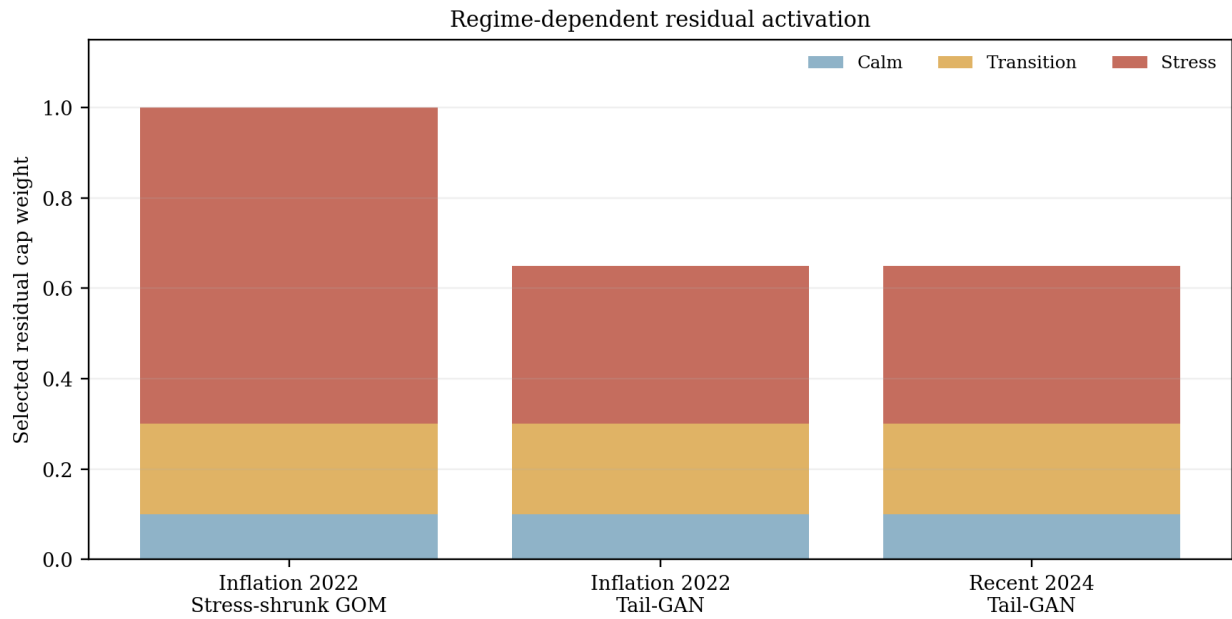


Figure 9. Regime-specific residual activation. GOM receives the largest stress allocation, while Tail-GAN is more tightly capped.

7.7 Risk calibration

The accepted models improve the average quantile score, but the 5% VaR exception process remains imperfect. In 2022, GOM records exception rates of 8.4% at one day and 9.3% at five days. The five-day unconditional-coverage p-value is approximately 0.008, and the independence p-value is near 10^{-9} . In 2024, Tail-GAN records a 7.9% exception rate at both horizons; the one-day independence test is close to the 5% boundary and the five-day test rejects independence strongly. The gated models therefore improve distributional scoring without eliminating clustered tail exceedances.

Table 6. Tail-risk diagnostics for accepted forecasts.

Fold	Model	Horizon	Exceptions	Kupiec p	Independence p	Pred. ES	Realized tail
Inflation 2022	Stress-shrunk GOM	1 day	8.4%	0.0315	0.58	-0.0312	-0.0299
Inflation 2022	Stress-shrunk GOM	5 day	9.3%	0.00782	1.11e-09	-0.0709	-0.0613
Inflation 2022	Tail-GAN	1 day	8.4%	0.0315	0.58	-0.0313	-0.0299
Inflation 2022	Tail-GAN	5 day	9.3%	0.00782	1.11e-09	-0.0710	-0.0613
Recent 2024	Tail-GAN	1 day	7.9%	0.0634	0.0461	-0.0145	-0.0156
Recent 2024	Tail-GAN	5 day	7.9%	0.0634	2.06e-10	-0.0296	-0.0296

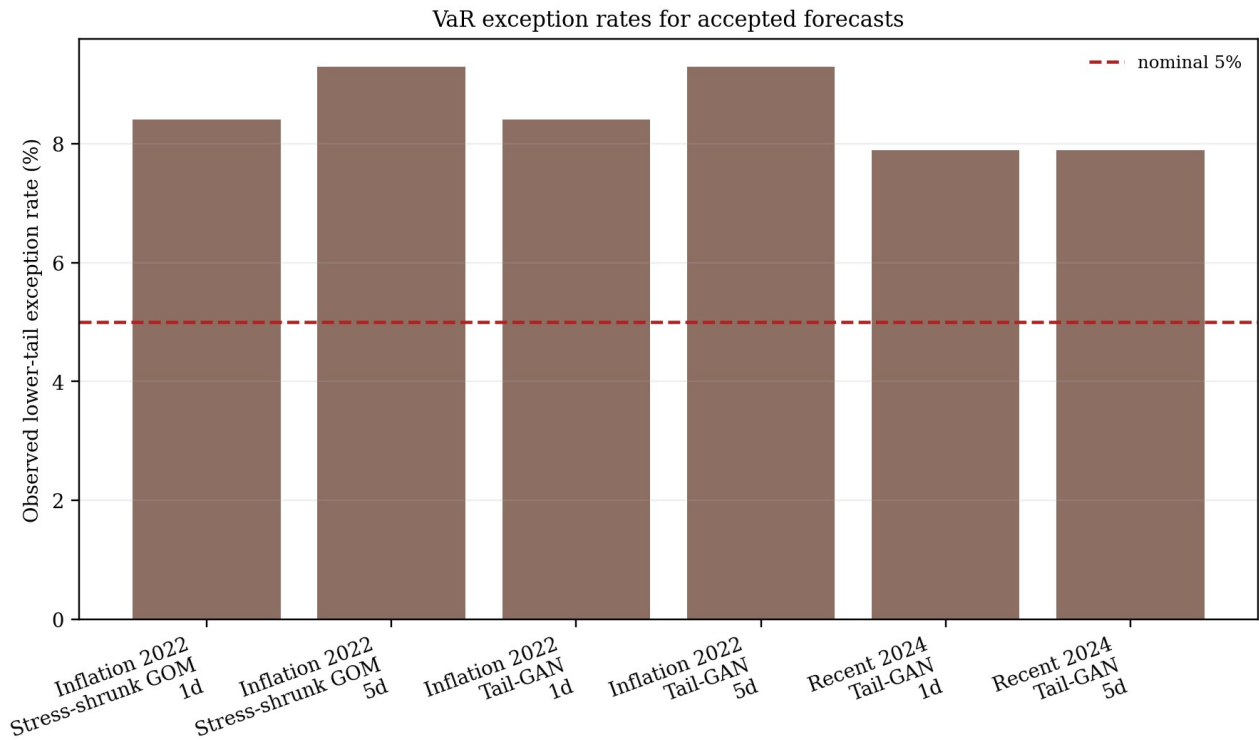


Figure 10. Observed lower-tail exception rates compared with the nominal 5% level.

8. Discussion

8.1 Source of the accepted gains

The 2022 improvements are not produced by wholesale replacement of the statistical model. They arise from small, confidence-weighted changes to the quantile surface, concentrated more strongly in the five-day horizon and in periods assigned higher transition or stress probability. This construction explains why a raw generator can be weaker than EWMA while the gated hybrid is stronger: the gate extracts a subset of the residual signal and suppresses the remainder.

GOM is most successful in the inflation fold because its stronger stress activation and tighter generator controls align with a regime characterized by persistent volatility and macroeconomic repricing. Tail-GAN is less specialized and is accepted both in 2022 and 2024, but its 2024 evidence is less precise. The structured Transformer remains close to the anchor in all folds. Its failure to pass the gate suggests that pretraining alone did not create a sufficiently stable incremental quantile signal.

8.2 Why the 2020 raw gains were rejected

The 2020 result exposes a central difficulty in regime forecasting. The crisis produced a large distributional break that was poorly represented in the preceding 2019 validation period. Raw scenario generators adapted favourably on the test sample, yet their validation-selection evidence did not identify a stable activation rule. The safety gate is intentionally asymmetric: it accepts a missed opportunity rather than a model whose apparent success becomes visible only after the test period is observed.

8.3 Statistical and practical significance

The absolute score changes are small, as expected when a strong conditional volatility model is the benchmark. The 2022 common-origin intervals support a persistent average improvement, whereas the 2024 interval does not exclude no difference. Even in 2022, the results should be interpreted as development-fold evidence rather than a universal statement about financial regimes. A single untouched holdout remains necessary to assess whether the promotion mechanism transfers beyond the folds that shaped the architecture.

8.4 Limitations

- Only three random seeds are used. Equal weights and $N_{\text{eff}} = 3$ remove concentration, but they do not fully characterize training variability.
- The target is SPY; cross-asset inputs support conditioning, but the study does not establish universality across instruments.
- The regime probabilities are engineered from causal stress features rather than estimated by a fully probabilistic latent-state model.
- The statistical champion never admitted the adaptive expert pool, so its additional complexity did not contribute to the reported gains.
- VaR exception clustering persists, especially at five days, indicating that aggregate quantile accuracy and dynamic tail calibration remain distinct objectives.

- Repeated architecture development on the three folds increases the importance of preserving the untouched 2025 evaluation.

9. Conclusion

A robust statistical benchmark and flexible machine-learning model need not be treated as mutually exclusive alternatives. In this experiment, the EWMA Student-t forecast provides the complete fallback distribution, while neural networks are allowed to contribute only a bounded residual whose activation is justified on earlier chronological data. Under that design, stress-shrunk GOM and Tail-GAN outperform EWMA in the 2022 inflation fold by 0.408% and 0.285%, respectively. Tail-GAN also produces a 0.203% improvement in 2024, although the descriptive test interval is inconclusive. No model is promoted in 2020.

The main empirical result is therefore conditional rather than universal: regime-gated generative residuals can improve a strong heavy-tail volatility benchmark in some market environments, and the 2022 gains survive both the declared gate and a favourable common-origin interval. The remaining challenge is dynamic tail calibration. The development specification is now sufficiently defined for a single evaluation on the reserved 2025 fold without further adjustment to the model families, thresholds or seed set.

Appendix A. Complete gate decisions

Table A1. Formal decision record.

Fold	Family	Accepted	Final gain	Bootstrap LB	N_eff	Decision reason
Crisis 2020	Patch Transformer	no	-0.000%	0.00e+00	3.0	selection improvement below threshold
Crisis 2020	Tail-GAN	no	-0.000%	1.44e-06	3.0	episode regime or confirmation rejected
Crisis 2020	Stress-shrunk GOM	no	-0.000%	-4.57e-07	3.0	bootstrap lower bound nonpositive
Inflation 2022	Patch Transformer	no	0.000%	0.00e+00	3.0	selection improvement below threshold
Inflation 2022	Tail-GAN	yes	0.285%	8.58e-06	3.0	multi episode bootstrap confirmed
Inflation 2022	Stress-shrunk GOM	yes	0.408%	1.04e-05	3.0	multi episode bootstrap confirmed
Recent 2024	Patch Transformer	no	0.000%	0.00e+00	3.0	selection improvement below threshold
Recent 2024	Tail-GAN	yes	0.203%	3.32e-06	3.0	multi episode bootstrap confirmed
Recent 2024	Stress-shrunk GOM	no	0.000%	0.00e+00	3.0	selection improvement below threshold

Appendix B. Portfolio-repository presentation

The public repository should contain source code, configuration files, tests, this paper, concise reproduction commands and a small set of summary figures. It should exclude raw market data, trained checkpoints, HPC logs, usernames, temporary paths and credentials. The README should lead with the research question, architecture diagram, three headline results and the exact command sequence for software tests and development-fold evaluation. A separate RESULTS.md file can distinguish accepted final forecasts from raw model scores and state that the 2025 holdout is not part of the published development table.

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